

Title: Conceptual Design using ER Model. Hotel reservation Customer data.

Steps involved in Creating ER Diagram:-

Step-1: Problem understanding & Requirement Analysis

Analyze the real world application: Hotel Management System.

Requirement

- Customer registration.

- Room reservation and availability.

Step-2:

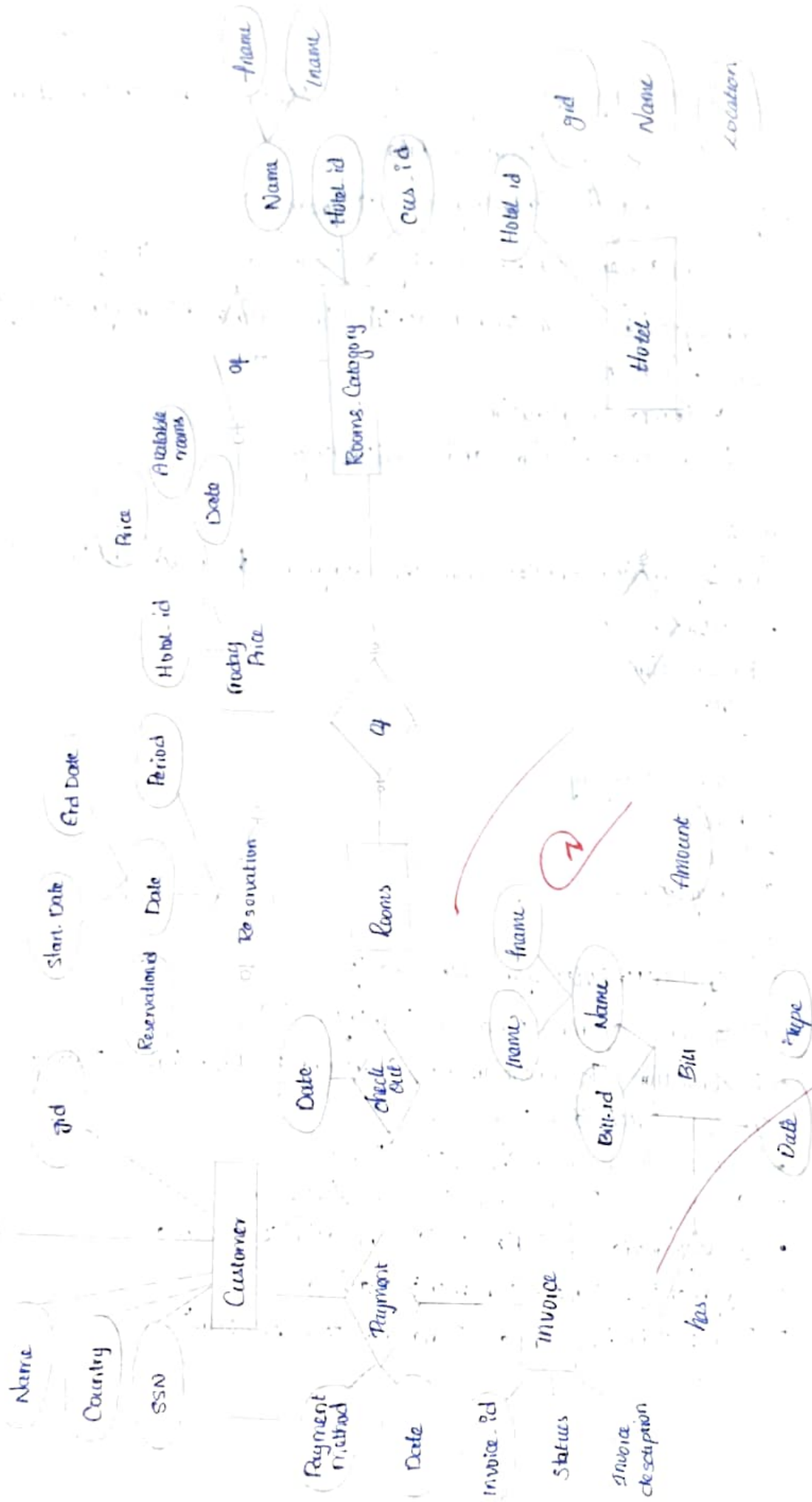
Identify major Entities from the diagram, the main entities are:

- Customer
- Reservation
- Hotel
- Room
- Room Category
- Invoice
- Bill
- Today price

Step-3

Identify Attributes for each entity.

- Customer: Name, Email, Country, SSN.
- Reservation: Reservation_id, Start date, End date, Date period.
- Hotel: Hotel_id, Name, Location.
- Room: Room_id, Check-Out, Date.
- Category: Room_id, Name, Hotel_id.
- Today price: Date, price, Available rooms.
- payment: Date, payment method.
- Bill: Bill_id, Date, Date, Type, Amount.



Step 4

Relationships Between Entities.

- Customer - Reservation

One-to-many (Customer can make many reservations)

- Reservation - Today price

- Reservation - Rooms :

One-to-many

- Room - Room Category

Many-to-one

- Room Category - Hotel.

Many-to-one

- Customer - payment

One-to-many

Step 5

Draw ER Diagram

- use rectangles for entities
- use Ovals for attributes
- use diamonds for relationships
- Link entities with attributes and relations using lines

VFLTECH	
EX 710	1
PERFORMANCE (S)	5
RESULT AND ANALYSIS (S)	5
VIVA VOCE (S)	5
RECORD (S)	5
TOTAL (20)	15
SIGN WITH DATE	20/7/05

Result :

This task helped us understand the importance of Conceptual design in database management. Using draw.io, we were able to visually model a real-time hotel management system into an ER diagram.

Customer	
Pk	Customer ID name Country Email

Reservation	
Pk	reservation ID Period Start Date Final Date

Today Price	
Pk	Hotel id price Rooms Available Date

Invoice	
Pk	invoicid Description Status

Room	
Pk	Roomid Hotel id Chelin Data Check Out data

Payment	
Pk	Payment ID
Pk	Reservation ID weekend Date

Bill	
Pk	Bill id france Date type Amount

Rooms categories	
Pk	Rooms Catege
Pk	Hotel id customerid Name

Hotel	
Pk	Hotel id Name gid taxes

1.2 Convert ER diagram into Relational model

Aim:

to design the relational model for Hotel Management System

Entity type represents table

All Single Value attribute become,

* Each entity in the ER diagram become the table

* the entity's attributes become the columns of the table
key attribute of the entity type represented by the primary key
the multi valued attribute is represented by a separate table

Composite attribute represented by components

partial attributes are not considered in the table

using these rules can convert the ER diagram to tables
and columns are assign the mapping between the tables

VEL TECH - CSE	
EX NO.	12
PERFORMANCE	5
RESULT	5
VIVA VO	3
REMARKS	15
TOTAL	25

25/125

VEL TECH - CSE	
EX NO.	12
PERFORMANCE	5
RESULT	5
VIVA VO	3
REMARKS	15
TOTAL	25

Result:

Thus, the ER diagram and Conversion of ER diagram to database relational model for Hotel management System has been implemented successfully.